Sara Kia

AI | Computer Vision | Physics | Software Engineering

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PROFESSIONAL SUMMARY

Physicist and AI professional with long-term experience in physics laboratory teaching, experimental work, scientific explanation, and technical project guidance. Background combines Physics, Artificial Intelligence, Python programming, image processing, and practical Computer Vision, with a strong interest in optics, electronics, measurement systems, and image-based analysis. Interested in applying AI, Python, OpenCV, analytical thinking, and measurement-oriented problem solving to automated quality analysis, scientific/technical imaging, and practical software engineering tasks. Currently strengthening software engineering foundations at 42 Heilbronn with C programming, Linux, Git, algorithms, and problem solving.

TECHNICAL SKILLS

Physics & Laboratory	Physics laboratory teaching, experimental demonstrations, measurement-oriented thinking, optics and electronics fundamentals, technical documentation, lab safety
Computer Vision & Imaging	OpenCV, image preprocessing, edge detection, thresholding, feature-based analysis, object detection basics, OCR basics, automated visual inspection concepts
Programming & Software	Python, C, SQL, basic Java, Linux, Git/GitHub, algorithms
Data & Machine Learning	Pandas, NumPy, Scikit-learn, data cleaning, data visualization, classification, evaluation metrics, statistics
Tools	VS Code, Jupyter Notebook, Google Colab, MATLAB, GitHub
Current Focus	Computer Vision portfolio, CVAT data labeling, YOLO object detection, C programming at 42 Heilbronn

PROFESSIONAL EXPERIENCE

Physics Assistant Labrotoruy | *Laser in Sacher Company , Marburg , Deutschland* | 2026–heute

- Conducting and evaluating physical measurements and experiments
- Setup, alignment, and testing of optical and laser-based measurement systems
- Performing measurements to characterize optical and electronic components
- Analysis and documentation of measurement results
- Supporting laboratory experiments, tests, and technical investigations
- Troubleshooting and optimization of measurement setups

Software Engineering Student - 42 Heilbronn | *2025 - Present* | Heilbronn, Germany

- Strengthening software engineering foundations through project-based learning in C programming, Linux, Git, algorithms, memory management, and peer-to-peer problem solving.
- Developing discipline in clean code, structured project work, and technical independence.

Freelance / Independent Technical Project Development | *2022 - Present* | Germany

- Worked independently on Python, data analysis, machine learning, and Computer Vision projects to rebuild a practical AI portfolio.
- Developed and documented projects involving image processing, face detection, OpenCV pipelines, and data-driven machine learning tasks.
- Currently developing a Computer Vision roadmap including CVAT labeling, YOLO object detection, and real-world inspection/detection projects.

Programmer / Technical Project Contributor - Amiri & Daraei Companies | *2013 - 2021* | Iran

- Worked on practical programming and technical problem-solving tasks in real-world company projects.
- Used programming, analytical thinking, and system design skills to support software development and internal technical workflows.
- Gained experience with Python, data processing, structured problem solving, and applied programming, forming a bridge between scientific background and later AI/computer vision work.

Robotics Instructor -Pajhoheshsaras 2013-2021

- Taught foundational robotics concepts, programming, and problem-solving to students; supervised hands-on robotics projects and competitions

Physics Laboratory Instructor / Physics Educator - University and School Laboratory Environments | *2000 - 2021* | Iran

- Worked full-time in physics education and laboratory environments, supporting practical physics learning, experiments, and scientific problem solving.
- Taught and explained physics concepts through experiments, demonstrations, laboratory exercises, and structured scientific reasoning.
- Guided students in observation, measurement, documentation, interpretation of results, and safe use of laboratory equipment.
- Developed strong skills in explaining technical topics clearly to different audiences, from students to colleagues.
- Built a strong foundation in analytical thinking, experimental accuracy, measurement-based reasoning, optics and electronics intuition, and clear technical communication.

SELECTED PROJECTS - AI, COMPUTER VISION & DATA

• **OpenCV Image Processing Pipeline:** Implemented core image processing techniques including grayscale conversion, thresholding, Gaussian blur, edge detection, resizing, cropping, and annotation.

• **Face Detection System:** Developed a real-time face detection project using Python, OpenCV, and Haar Cascade classifiers with bounding boxes and image/video processing concepts.

• **License Plate Detection:** Developed a computer vision pipeline for detecting and extracting vehicle license plates using preprocessing and feature-based techniques.

• **Persian Handwriting Recognition:** Worked on image preprocessing and classification methods for recognizing Persian handwritten text.

• **SpaceX Launch Prediction:** Built a machine learning classification project using real-world structured data, including data cleaning, feature engineering, model evaluation, and interpretation.

SLAM Robot Navigation: Worked on a Simultaneous Localization and Mapping system, using sensor fusion of LiDAR and ultrasonic data to build environment maps while tracking the robot's position in real time; combined both sensors to compensate for LiDAR's limitations on transparent and reflective surfaces.

EDUCATION

Software Engineering Program - 42 Heilbronn | *2025 - Present*

M.Sc. Artificial Intelligence - Amirkabir University of Technology |

B.Sc. Physics - University of Tehran |

LANGUAGES & STRENGTHS

Languages: Persian - Native | English - Fluent | German -Fluent

Strengths: Analytical Thinking | Scientific Communication | Problem Solving | Structured Working Style | Fast Learning | Reliability | Teamwork